Bruce Peters

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EXPERIENCE

H2X Lab, Boston University (RISE) – Research Intern, *PI: Prof. Eshed Ohn-Bar*

JUNE 2026 - PRESENT

- Built a multimodal pipeline that detects off-task behavior in small-group phonics tutoring videos, using pose, head pose, gaze, scene objects
- Benchmarked six models across three approaches (tabular TabPFN/XGBoost, a frozen DINOv2 encoder with a trainable head, a pose graph network) plus a zero-shot VLM (Qwen3-VL), with leave-one-session-out cross-validation on 60K hand-coded windows
- Showed a frozen DINOv2 encoder with a small trained head beat the best hand-engineered model
- Wrote the on- and off-task annotation taxonomy and trained additional annotators
- Paper "Pretrained Self-Supervised Visual Representations for Off-Task Detection in Early Reading Instruction," submitted to LAK 2027

Iron Panthers Robotics, Burlingame — *Programming Manager (2025 - present); Programming Training Lead (2024-2025)*

OCTOBER 2023 - PRESENT

- Wrote and maintained production Java for the robot that won the 2025 FRC World Championship (3,700+ teams), coordinating an elevator and pivot superstructure with safety interlocks while under time restrictions and competition pressure
- Built a custom WPILib robot-simulation platform that tripled programming-team efficiency by eliminating waits for physical robot time
- Designed and taught a Java training curriculum for new programmers, improving onboarding and code quality across the team
- Built a React-based scouting web application used by the full team during competitions for real-time data analysis
- Contributed to autonomous routines and computer vision pipelines using OpenCV and PhotonLib
- Collaborated within a 70+ member team under strict 6-week build deadlines

SKILLS

Languages: Python, Java, C#, C++, PHP, JavaScript, TypeScript, HTML/CSS

Frameworks & Libraries: React, Firebase, FastAPI, OpenCV, TensorFlow, PyTorch

Tools & Platforms: Git/GitHub, AWS (EC2, RDS), Vite, VS Code

Technical Concepts: Object-Oriented Programming, ML model training, REST APIs, Computer Vision

AWARDS

FRC World Championship Winner (2025)

FIRST Leadership Award (formerly Dean's List) Semi-Finalist (2026)

2nd Place Congressional App Challenge (CA-15), Word Wiz AI

1st Place Panther Hacks Hackathon

1st Place LG Hacks 2.0 Hackathon

2nd Place UHS Hackathon (CanvAI)

PUBLICATIONS

LAK 2027 (submitted): Pretrained Self-Supervised Visual Representations for Off-Task Detection in Early Reading Instruction

NeurIPS 2026 On-Device Intelligence Workshop

Burlingame High School, Burlingame — *Student Body President* (2026-present); *Student Body Treasurer (2025-2026)*

APRIL 2025 - PRESENT

- Elected student body president of ~1,600 students
- Managed a ~$70K student government budget and tracked allocations for major school events
- Coordinated with a 4-member executive cabinet to plan school-wide events, including Homecoming
- Planned Homecoming to 1,026 tickets sold (a school record), netting ~$30K profit on ~$10K in expenses to fund the next year's leadership program
- Oversaw budgeting, accounting, and communication within the leadership elective

PROJECTS

Word Wiz AI — *AI-Powered Reading Tutor*

- Built an AI-driven educational web app that provides phoneme-level pronunciation feedback for early readers (280+ users , 13,000+ organic visitors)
- Implemented grapheme-to-phoneme alignment and speech analysis to identify pronunciation errors and cut response latency from 15 seconds down to 2 seconds
- Developed a full-stack system using React, Flask, Firebase, and ML models for personalized phonics practice
- Awarded 2nd Place in the Congressional App Challenge (CA-15)

Crab Hunt — *Panther Hacks (1st Place)*

- Built a community building scavenger hunt called Crab Hunt
- Combined a mobile app with a physical prize box to introduce a large variety of puzzles for users as they solve the scavenger hunt
- The app used AI to parse information about the users (that they provided in the form of short answers before starting the activity) to create custom clues and questions that forced the users to learn more about each other.

Pill Pal — *LG Hacks 2.0 (1st Place)*

- Built an AI-assisted smart pillbox with a web app, voice interface, and motorized hardware controls
- Integrated physical hardware with an AI assistant for medication guidance and symptom logging

CanvAI — *UHS Hackathon (2nd Place)*

- Built an AI-powered assistant for the popular school LMS Canvas
- Used vector representations of assignments to quickly search and

(accepted, co-first author): Compression Has an Accent

NHSJS (in revision): Phoneme Level Pronunciation Feedback for Early Literacy

sort tasks based on importance and difficulty

EDUCATION

Burlingame High School, Burlingame

AUGUST 2023 - PRESENT

Relevant Coursework: AP Computer Science, AP Physics, AP Calculus BC, APUSH

College of San Mateo — *Concurrent Enrollment*

JANUARY 2025 - AUGUST 2025

Relevant Coursework: Python Programming, Object-Oriented Program Design, Programming Methods in Java, Programming Methods in C++, PHP Programming